Fair and Load-Balanced Student-Alumni Matching at VJTI

Multi-Objective Capacitated Bipartite Assignment with Rawlsian Fairness & Departmental Parity

Course: Optimization Techniques (University Group Project)
Institution: Veermata Jijabai Technological Institute (VJTI), Mumbai
Notice: ARTIFICIAL - FOR TESTING ONLY (All benchmark data utilize pseudonymous IDs: S001, A001)

Slide 1: The Problem & Institutional Context

The Challenge at VJTI:

- Each year, hundreds of graduating engineering students across **8 academic departments** seek alumni mentorship.
- **Traditional matching** relies on first-come-first-served or greedy score maximization.

Critical Failures of Greedy Approaches:

1. **Mentor Burnout**: Top alumni receive overwhelming request loads, leading to dropouts.
2. **Peripheral Student Neglect**: Students with average GPAs or cross-domain aspirations receive poor matches.
3. **Departmental Inequity**: Computing branches (CE/IT) dominate high-scoring alumni; core branches (Production, Textile, Civil) face structural deficits.

Slide 2: Mathematical Formulation

Decision Variables:

- $x_{ij} \in {0, 1}$: Binary match between student i and alumnus j.
- $y_j \in {0, 1}$: Alumni activation status ($\sum_i x_{ij} \ge 1$).

The Tri-Objective Trade-off:

$$\min \mathbf{F}(\mathbf{x}) = \begin{bmatrix} -Z_1 \ -Z_2 \ +Z_3 \end{bmatrix}$$

1. **$\max Z_1 = \sum_{i,j} q_{ij} x_{ij}$**: Utilitarian Match Quality (Total Human Capital).
2. **$\max Z_2 = \min_i Q_i$**: Rawlsian Egalitarian Fairness (Protect the least-advantaged student).
3. **$\min Z_3 = \max_j L_j$**: Workload Balancing (Prevent mentor overload).

Slide 3: Operational Constraints

Constraint	Mathematical Form	Operational Purpose
Degree (K)	$\sum_j x_{ij} = K, \forall i$	Exactly K mentors per student ($K=2$)
Capacity (C_j)	$\sum_i x_{ij} \le C_j y_j, \forall j$	Mentor cannot exceed capacity cap
Consent (c_j)	$x_{ij} \le c_j, \forall i,j$	Prior opt-in mandatory
Threshold ($q_{\min}$)	$x_{ij} = 0 \text{ if } q_{ij} < q_{\min}$	No low-quality matches allowed
Quality Floor (θ)	$Q_i = \sum_j q_{ij} x_{ij} \ge \theta$	Guarantees minimum mentorship quality
Branch Parity (δ)	$	\overline{Q}*b* - *overline{Q}*{b'}
Diversity Cap	$\sum_{j \in \text{Comp}(k)} x_{ij} \le 1$	Prevent single-firm clustering

Slide 4: Algorithmic Architecture

Our methodology benchmarks three methodological tiers:



Slide 5: Custom Feasibility Repair Operator

Why Generic Genetic Operators Fail:

- Discrete integer chromosomes $X \in [0, M-1]^{N \times K}$ easily produce duplicate alumni, capacity overflows, and diversity violations during crossover/mutation.

Two-Pass MatchingRepair Mechanism:

- **Pass 1 (Student Level)**:
- Filters non-consenting alumni ($c_j = 0$) and sub-threshold scores ($q_{ij} < q_{\min}$).
- Enforces corporate diversity cap (at most 1 alumnus per company).
- Fills shortfalls greedily from sorted feasible candidate pools.
- **Pass 2 (System Level Load Re-balancing)**:
- Identifies bottleneck alumni ($L_j > C_j$).
- Systematically swaps students from overloaded mentors to underutilized candidates with highest available q_{ij}, preserving quality floors and branch parity.

Slide 6: Small Instance Benchmark (50×30)

True Pareto Frontier traced by HiGHS MILP vs Evolutionary Solvers:

Algorithm	Total Quality Z_1	Worst Student Z_2	Max Load Z_3	Hypervolume	Spacing	Runtime (s)
Random Baseline	45.95	0.60	8	—	—	0.012
Greedy Baseline	72.62	1.02	8	—	—	0.002
Min-Cost Flow	73.35	1.02	8	—	—	0.053
Exact MILP	[62.4 - 73.3]	[1.02 - 1.25]	[4 - 8]	**443.98**	2.25	14.25
NSGA-II	[68.1 - 70.4]	[0.98 - 1.12]	[6 - 7]	115.75	0.01	3.05
SPEA2	[68.0 - 69.8]	[0.99 - 1.10]	[6 - 7]	118.29	0.00	3.21
Hybrid Memetic	[69.2 - 72.8]	[1.02 - 1.21]	[5 - 7]	**232.17**	0.15	**2.96**

Observation: Hybrid Memetic NSGA-II doubles the Hypervolume of standard NSGA-II and discovers high-quality compromise solutions in under 3 seconds.

Slide 7: Scalability Across Cohort Dimensions

Benchmarking across Small (50×30), Medium (300×150), and Large (1000×500):

Scale	Demand ($N \times K$)	Greedy (s)	Min-Cost Flow (s)	NSGA-II (s)	Hybrid Memetic (s)
Small (50x30)	100	0.007	0.133	4.01	4.34
Medium (300x150)	600	0.126	0.969	26.11	19.79
Large (1000x500)	2000	1.248	13.03	61.71	92.53

- **Hypervolume Performance at Large Scale**:
- Raw NSGA-II: $HV = 845.95$, Gini = 0.152
- **Hybrid Memetic NSGA-II**: **$HV = 2297.35$ (+171.6%)**, **Gini = 0.0435 (3.5x lower inequality)**

Slide 8: The Price of Branch Parity (δ)

How much total efficiency does VJTI sacrifice to guarantee equal departmental outcomes?

Disparity Tolerance δ	Max Quality Z_1	Achieved Disparity	Efficiency Impact
0.15 (Ultra-Strict)	65.80	0.148	-9.0%
0.25 (Balanced Parity)	69.47	0.244	-3.9%
0.35 (Standard Parity)	72.32	0.326	Reference (0%)
0.50 (Relaxed Parity)	70.61	0.428	-2.4%

Strategic Recommendation: $\delta = 0.25$ provides an ideal sweet spot, eliminating perceived bias against core engineering departments at an efficiency cost of only 3.9%.

Slide 9: Secondary Scenario: Minimizing Active Alumni

Can we achieve top-tier student matching while contacting fewer alumni?

- **At $\ge 80\%$ Max Quality**: Only **15 alumni** needed ($H = 2.75$)
- **At $\ge 90\%$ Max Quality**: Still only **15 alumni** needed ($H = 2.60$)
- **At $\ge 95\%$ Max Quality**: **16 alumni** required ($H = 2.85$)

Administrative Value: VJTI can achieve 90% of maximum mentorship value by engaging just **50% of its alumni roster**, saving significant volunteer coordination effort while maintaining high company entropy.

Slide 10: Conclusion & Implementation Roadmap

Key Findings:

1. **Hybrid Metaheuristics Excel**: Seeded Memetic NSGA-II consistently dominates standard MOEAs in Hypervolume and fairness.
2. **Equitable Allocations**: Gini inequality dropped to 0.043, proving high equity across all students.
3. **Scalability**: Solves 1,000 students x 500 alumni in <95 seconds.

Roadmap for VJTI:

- **Phase 1**: Embed Seeded Memetic NSGA-II engine into the VJTI Alumni Portal backend.
- **Phase 2**: Standardize departmental disparity policy at $\delta = 0.25$.
- **Phase 3**: Enforce employer diversity cap to ensure multifaceted career mentorship.